

ESP32-C3 Zero Protective Case

Product Slideshow Template | Student name | Student number

Product Overview

Product name: ESP32C3 Zero Protective Case

Purpose: Protect the ESP32-C3 Zero board during classroom use, maintaining access to USB-C, GPIO pins, and buttons.

Problem solved: Boards are frequently handled and are at risk of short circuits, bent pins, and breakage from drops. The case protects the board while still leaving the USB-C port, GPIO pins, and BOOT/RST buttons accessible. Ventilation is provided by holes on both sides and the lid (No. 10 square holes 4mm^2 in two rows on each side and No. 5 square holes 4mm^2 on the lid). Model created in Tinkercad. Printed in PLA with a BambuLab A1 printer. The product was developed in three prototyping phases, with testing and improvements at each iteration.

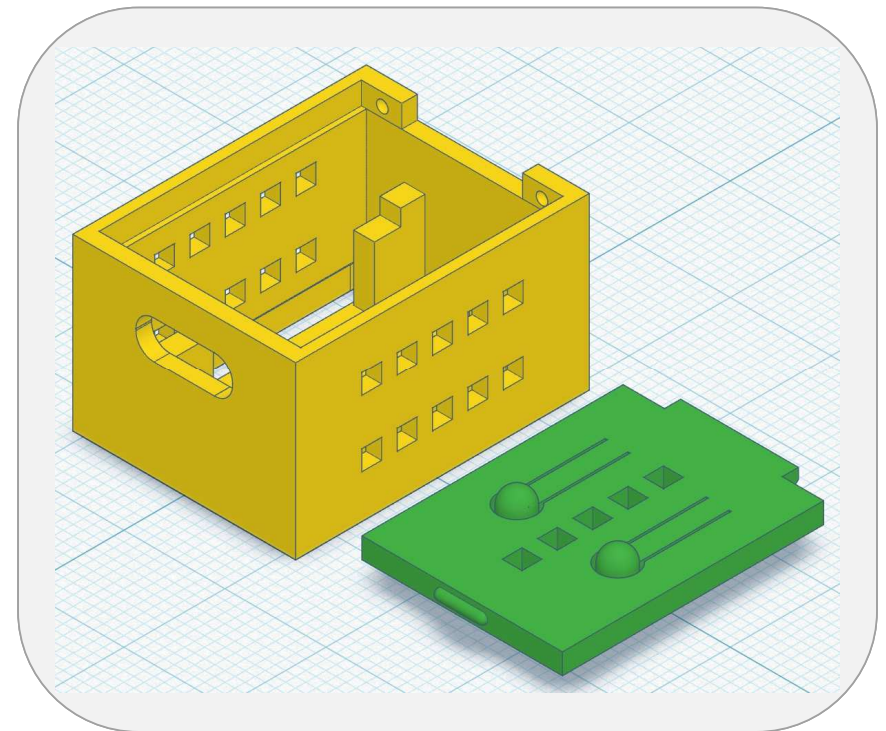
Intended users: Students, makers, and engineers who use microcontrollers in the classroom or lab.



Key Features

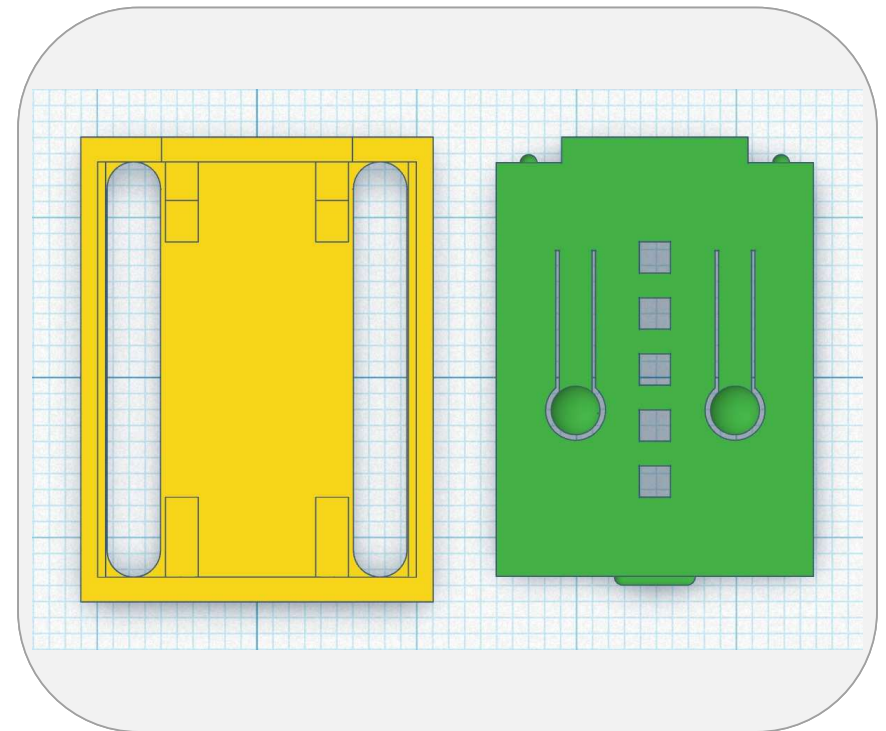
ESP32-C3 Zero board and component protection.
Ventilation on the sides and cover (No. 10 square holes 4mm² in two rows on each side and No. 5 square holes 4mm² on the lid).

Access to USB-C, GPIO pins, and BOOT/RST buttons.
Safe and precise housing for the ESP32-C3 Zero.



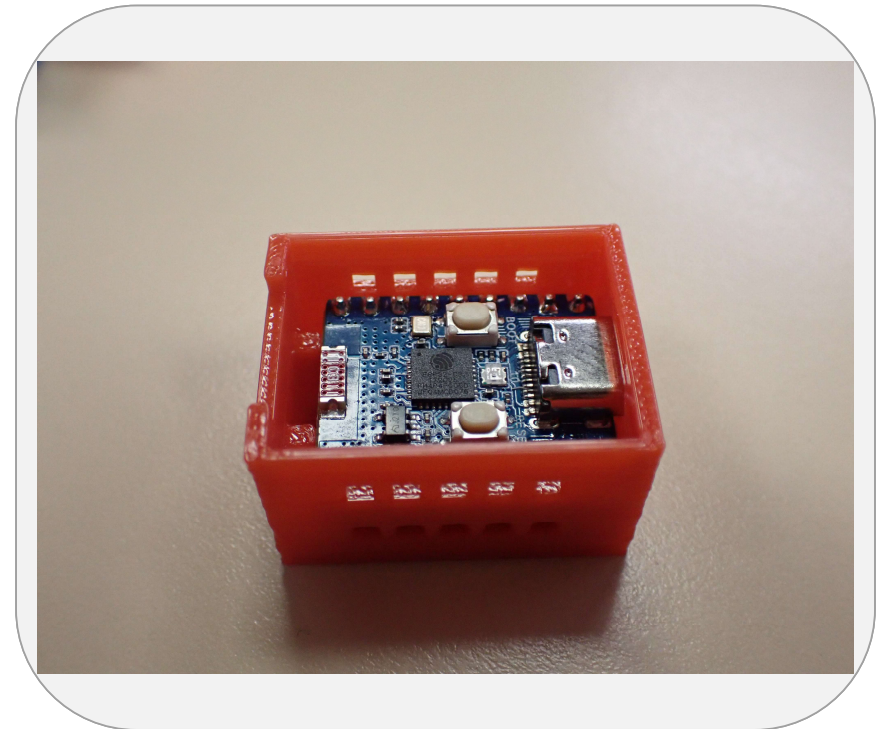
Design and Prototype Images

Top view of the cover (green) and base (yellow)
Cover:
Two flexible buttons and ventilation holes
Base: Side slot for the GPIO pins.



Testing and Improvements

Tested: Insertion of the ESP32-C3 Zero board, access to USB-C, pins and buttons, ventilation, robustness, and print quality. **Prototype 1 issues:** Internal supports too large (card won't fit); pins protruding from the bottom; cover too narrow (warped wall); button pins too long (buttons getting stuck); insufficient ventilation. **Modifications (Prototype 2):** Supports widened to accommodate the board; bottom deepened to protect the pins; cover refined; button pins shortened; more ventilation holes on the sides. **Remaining issues:** Pins still close to the bottom, slots not centered, ventilation needs to be centered, cover still a bit tight. **Final design:** Centered slots; raised bottom (pins 0.1 mm from the surface); more ventilation holes on the covers and centered (the lowest ones on the sides lowered by mm to fit under the board); optimized cover tolerance.

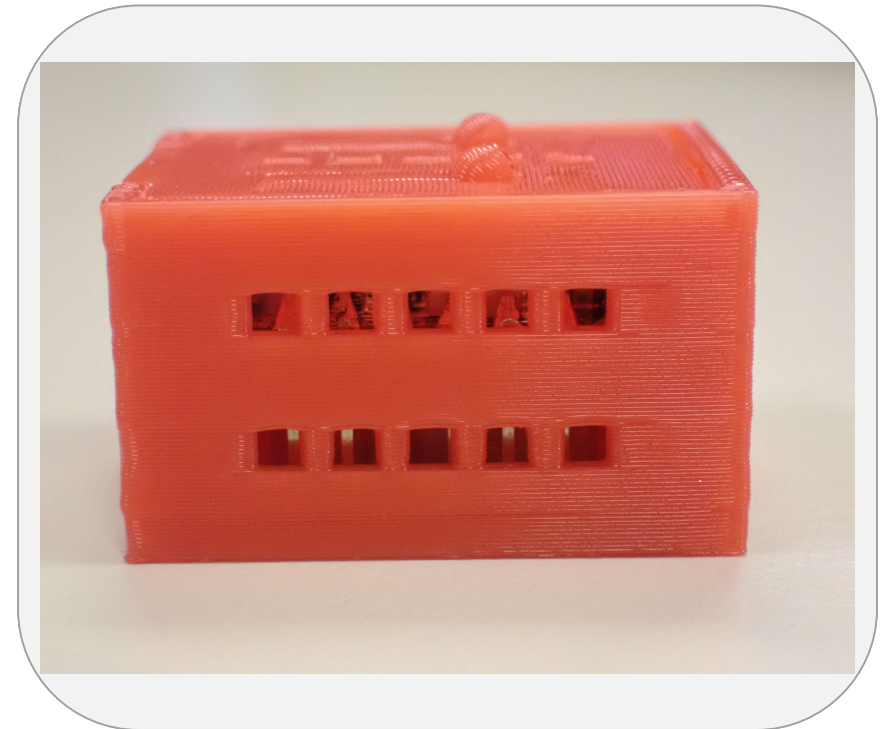


Benefits of the Design

The ESP32-C3 Zero board is protected from shock, dust, and accidental contact.

The USB-C port is accessible, and the buttons can be operated from the outside when the case is closed. This allows the board to be transported and used safely in the classroom.

Final version: adequate ventilation, precise fit, and robust construction.



Final Reflection

What worked well: The iterative process allowed me to correct every problem; the flexible buttons are an effective solution.

What I would change: Optimize the wall thickness sooner and test the locking clips earlier.

What I learned: The importance of measuring accurately, testing real prototypes, and using feedback to improve the design.